

Automated daily report generation from construction videos using ChatGPT and computer vision

Bo Xiao^a, Yifan Wang^a, Yongpan Zhang^b, Chen Chen^{c,*}, Amos Darko^d

^a Department of Civil, Environmental, and Geospatial Engineering, Michigan Technological University, Houghton, 49931, MI, USA

^b Department of Building and Real Estate, The Hong Kong Polytechnic University, Hung Hom, Kowloon, Hong Kong, China

^c School of Civil Engineering and Architecture, Zhejiang University of Science and Technology, Hangzhou 310023, China

^d Department of Construction Management, University of Washington, Seattle, 98195, WA, USA

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ABSTRACT

Daily reports are important in construction management, informing project teams about status, enabling timely resolutions of delays and budget issues, and serving as official records for disputes and litigation. However, current practices are manual and time-consuming, requiring engineers to physically visit sites for observations. To fill this gap, this paper proposes an automated framework to generate daily construction reports from on-site videos by integrating ChatGPT and computer vision (CV)-based methods. The framework utilizes CV methods to analyze video footage and extract relevant productivity and activity information, which is then fed into ChatGPT using proper prompts to generate daily reports. A web application is developed to implement and validate the framework on a real construction site in Hong Kong, generating daily reports over a month. This research enhances construction management by significantly reducing documentation efforts through generative artificial intelligence, with potential applications in jobsite safety management, quality reporting, and stakeholder communication.

1. Introduction

In construction management, daily reports are one of the most critical documents for recording construction progress, schedules, quality of works, and daily affairs [1,2]. Daily reports are important in construction projects for 1) keeping the project management team updated with the project status to solve any delays/over-budgeting issues on-time [3], 2) allowing general contractors to examine project performance across all areas of the projects [4], and 3) serving as official documents for post-project disputes or even evidence in lawsuits. In all types of construction projects, engineers need to write daily reports by physically visiting their sites and conducting manual observations several times per day [1,5]. Considering modern construction projects are huge and complex, engineers need to expend significant efforts to collect and summarize such comprehensive project-related information. Automating the generation of daily reports via high-tech solutions is especially meaningful for construction management because it saves construction engineers substantial manual efforts, allowing them to focus on engineering works instead of documentation.

Construction video captured using on-site cameras contain visual information on project activities, progress, and status, which can help engineers to remotely monitor their sites and then write reports [6]. However, Xiao et al. [1] has indicated that construction engineers are quickly fatigued by watching construction videos. Recently, computer vision (CV) methods have been introduced in construction management to analyze on-site videos automatically, with reliable performance being achieved in many tasks, including productivity analysis, safety control, and progress monitoring [7]. For example, Chen et al. [8] proposed a framework for recognizing construction machine activities and calculating the earthmoving productivity of this equipment. To facilitate construction safety, Kim et al. [9] proposed a vision-based safety assessment method for monitoring struck-by accidents in construction sites.

However, an integrated framework for applying CV techniques to automatically generate daily reports is lacking in construction management. Currently, vision-based methods can extract specific project-related information (e.g., productivity, progress, and safety) from construction videos. However, how to present this fragmented information

* Corresponding author.

E-mail addresses: boxiao@mtu.edu (B. Xiao), wyifan@mtu.edu (Y. Wang), yongpan.zhang@polyu.edu.hk (Y. Zhang), chen.c@zust.edu.cn (C. Chen), adarko@uw.edu (A. Darko).

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0926-5805/© 2024 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

in the format of natural sentence descriptions, like reports written by humans, to help construction managers and stakeholders understand the project information has not been elucidated. Additionally, construction reports must include insights from construction engineers to analyze daily activities and provide recommendations based on their professional knowledge of construction management. Currently, vision-based methods have limited reasoning ability to analyze project delays/over-budgeting, which makes it difficult to provide engineering insights for daily report generation using this technology.

The recent development of large language models (LLMs) in artificial intelligence (AI) has created new opportunities for the generation of construction reports. LLMs are natural language processing techniques for training deep learning models (i.e. transformer architecture) on vast amounts of textual data from the Internet, enabling them to generate human-like language. The Chat Generative Pre-training Transformer (ChatGPT) [10] is a representative LLM launched by OpenAI in November 2022, which has attracted widespread interest from society. As a generative AI, ChatGPT's ability to understand and respond to human language, as well as its vast knowledge and expertise, accuracy and reliability, and speed and efficiency have made deep impressions on the public [10,11]. ChatGPT [10] also has immense application potential in construction management in terms of automating construction project data analysis, facilitating communication between stakeholders, and generating daily reports.

The objective of this research is to propose a novel framework for generating daily reports from construction videos by integrating ChatGPT and CV. The proposed framework adopts vision-based object detection, tracking, and activity recognition methods to extract productivity information from construction sites. Then, the productivity information is formalized by prompt engineering and inputted to ChatGPT to generate reports. In addition, a web application is developed to implement the proposed framework, which allows construction engineers to stream onsite videos using Internet Protocol (IP) cameras, visualize project productivity information, and browse daily reports. The ability of the proposed framework to generate daily reports is tested over one month in a real construction site in Hong Kong. This research contributes to construction management by saving construction professionals the effort of daily project documentation by leveraging construction videos and AI. By fine-tuning the vision modules, the proposed framework can be applied to all types of construction projects (e.g., residential, industrial, and infrastructure projects).

2. Literature review

In this section, existing research on CV and language processing (NLP) in construction management is reviewed. The development of GPT methods is then comprehensively introduced. Finally, the research gaps and objectives in this section are recognized.

2.1. Vision-based monitoring in construction

Vision-based monitoring has been widely used for automatic construction management in recent years for various tasks, such as safety control, operation process monitoring, and scene analysis [12–14]. Some researchers utilized object detection to identify construction objects in real time for safety control. For example, Ferdous and Ahsan [15] created an anchor-free you only look once (YOLO) network to detect hardhats, vests, and safety glasses for worker safety management. Li [14] proposed a deep convolutional neural network (CNN) to detect falling objects on construction sites to protect construction workers from injuries. Some researchers combined object detection with a tracking method to monitor the operation of construction machines. Xiao and Kang [16] developed a construction machine tracker system, which combined a YOLOv3 detector and an association-based tracking algorithm to monitor the trajectories of construction machines. The proposed method achieved an accuracy of 96.9 % in tracking multiple

construction machines in the video.

Activity recognition is another vision-based monitoring technique in construction. Researchers applied spatial-temporal-based CNNs to recognize activities (e.g., working, idling, and digging) of construction machines (e.g., excavator and truck) and calculate productivity based on the activity information [17,18]. Some research work focused on recognizing the activity of workers to monitor their work efficiency. For example, Luo et al. [19] used two-stream CNN to recognize 16 categories of workers' activities (e.g., fixing rebar, placing framework, and moving). Then, by combining the activity context with deep features, Luo et al. [20] discovered relations among people and recognized the work groups of construction workers.

Scene analysis applies a variety of vision-based methods to identify additional information from construction images and videos, which helps to understand complex construction scenes. Kim et al. [21] used a semantic graph neural network and deep visual features to structuralize construction resources and analyze the relationships between construction resources (e.g., worker, machine, and material). Wang et al. [2] integrated semantic information extraction, object detection, and image caption methods to describe a construction scene using a sentence. Xiao et al. [12] combined a YOLOv3 detector, Kalman filter-based tracker, and data augmentation algorithm to identify construction machines during nighttime to avoid accidents caused by poor light conditions. In addition, Chen et al. [22] combined machine detection, tracking, and activity recognition results with a visual distance calculation algorithm to identify the work group of excavators and trucks and analyze the reasons for trucks idling.

The above reviews indicated that current research mainly focused on extracting individual project-related information from construction videos, such as object detection (e.g., worker, machinery) [12–16] and activity recognition [17–20]. Moreover, existing scene analysis methods [2,12,21,22] simply summarize construction scenes with words or sentences. Such information provides only basic details, such as the number and type of workers or machinery present, which must then be manually integrated and analyzed to generate useful productivity insights for construction managers requiring additional human efforts. Therefore, an integrated framework is needed to systematically organize useful visual information and deliver productivity reports in natural language, offering construction managers more direct and easily interpretable insights.

2.2. Natural language processing in construction

Construction documents are strongly associated with human language and contain important project-related information, which cover a full range of construction phases. Recent NLP methods have the potential to greatly improve the efficiency and quality of reviewing, comprehending, classifying, and retrieving knowledge from documents [23]. In recent years, NLP has achieved promising results in different construction management tasks, such as safety control, contract analysis, staff arrangement, and infrastructure management.

Safety control by knowledge retrieval and classification is one of the main NLP application tasks. Tixier et al. [24] designed a rule-based NLP system to extract and organize construction injuries from unstructured reports. Chokor et al. [25] applied K-means clustering to classify safety reports and analyze the distribution of construction accidents in the State of Arizona. Zhang et al. [26] developed an NLP model with support vector machine (SVM), linear regression (LR), K-nearest neighbor (KNN), decision tree (DT), Naive Bayes (NB), and sequential quadratic programming (SQP) algorithms to analyze the accident report. They classified accidents into 11 types and analyzed the reasons for the distribution of accidents. However, Luo et al. [27] indicated that these ML-based methods are not efficient to use, which need to manually select effective text features. More efficient deep learning-based methods should be created to automatically learn text features.

NLP is also widely used to address legal and contractual problems by

compliance checking. For example, Salama and El-Gohary [28] first proposed an ML-based NLP model to classify the regulatory provisions into 14 categories (e.g. environmental, safety, quality, etc.) for compliance checking. Zhang and El-Gohary [29] proposed a rule-based semantic NLP approach to extract regulatory requirements from construction documents and transformed them into a formalized format that enables automated reasoning. Song et al. [30] developed a DL-based method to analyze the building regulations. The proposed method used word2vec model for the semantic analysis of words. Zhong et al. [31] indicated that aforementioned DL-based methods can only learn shallow semantics information, which is limited for text analysis. A method should be created to analyze the logistic relationships between construction regulations. Therefore, they [31] proposed a hybrid deep neural network, which combined the bidirectional long short-term memory (Bi-LSTM) and the conditional random field (CRF) models. The proposed Bi-LSTM-CRF can automatically extract construction procedural constraints from construction regulations for both entities and interlinks or relationships between them. Typically, Bi-LSTM is used for pattern recognition, which can learn the relationships between different entities in regulations.

To address disputes, Moon et al. [32] developed a bidirectional encoder representation from transformer (BERT)-based model [33] for contractual risk classification. The contractual risk categories were classified based on the construction specification clauses to provide reviewers with the crucial clauses that commonly cause disputes. BERT [33] is a state-of-the-art pretrained NLP model composed of bidirectional transformer encoder blocks. The model was pretrained from the massive amounts of language data and achieved stratifying performance within minor fine-tuning. Moreover, the bidirectional structure enables easy and deep understanding of the text, as well as the relationship between sentences and words. Kim et al. [34] also applied BERT to retrieve infrastructure damage information in terms of question answering. They first used a sentence-BERT model to retrieve question and answer (QA) sentences about infrastructure damage from published reports and created several QA datasets. Then, a BERT model was trained with the question-answer pairs in the QA datasets. Finally, additional 142 question-answer pairs created from the same reports were used for testing, with an accuracy of 90.5 % being achieved.

Currently, the NLP techniques have been applied in construction documents and contract analysis, but still require significant human efforts [24–33]. For example, ML-based require manual selection of effective text features [24–27], and DL-based methods need major or minor training on large dataset [28–33]. Moreover, current NLP methods only provide language information as simple words or sentences. Therefore, there is a clear need for a more efficient approach to construction report generation that can deliver comprehensive and integrated construction text information.

2.3. Development of GPT-based LLMs

GPT series LLMs were developed by OpenAI and can generate natural language text based on the given input. GPT-1 [35] was first proposed in 2018, which aimed to learn a universal representation that transfers to a wide range of tasks with minor adaption. The training processes of GPT-1 have two stages, first, large unlabeled data were used to train a high-capacity language model with unsupervised learning. Subsequently, the model was fine-tuned and transferred with labeled data to a target task with supervised learning. In the first stage, a 12-layer decoder-only transformer with a masked self-attention model was used for handling long-term dependencies in text. The model was then fine-tuned for a variety of supervised tasks for natural language inference, question answering, semantic similarity, and text classification. GPT-1 was a significant breakthrough in the field of natural language processing, and it paved the way for the development of more advanced language models, such as GPT-2 and GPT-3.

GPT-2 [36], which is an improved version of GPT-1, was released in

2019 and trained on a larger corpus of data (40 GB). Specifically, four versions of GPT-2 were released by OpenAI, with model sizes ranging from small (124 million parameters) to large (1.5 billion parameters). The number of layers in the decoder-only transformer model was increased to 48. GPT-2 was trained on several datasets with multiple tasks, and it did not need fine-tuning with supervised learning for a specific task. According to the researchers, if a language model is trained on sufficiently large and diverse datasets, it can perform well across various tasks without further supervised fine-tuning. Furthermore, human correction was used for data filtering (i.e., manually filtering some of the documents), which enables GPT-2 to generate outputs credible to humans [37]. The performance of GPT-2 was superior to seven out of eight tested state-of-the-art methods. However, the performance of GPT-2 in summarization and translation tasks was worse than those of the best methods [38].

GPT-3 was improved based on the GPT-2 model, which used a transformer model with 96 layers and 175 billion parameters. GPT-3 performs well in tasks such as translation, question answering, cloze filling, new word usage, and algebraic operations, but its performance on reading comprehension and reasoning task datasets, as well as long test generation, still needs to be improved. ChatGPT is a chatbot model developed from GPT-3, which can quickly generate detailed responses and articulate answers conversationally. ChatGPT applies reinforcement learning from human feedback to fine-tune GPT-3; thus, it can follow a broad class of written instructions (uses human-preferences as a reward signal to fine-tune models) [37]. In the training stage, hired labelers first provided demonstrations of the desired output of the question in the dataset. These data were used to fine-tune GPT-3 with supervised learning to train a supervised policy. A comparison dataset was then created, in which labelers indicated model outputs that they preferred for a given input. Accordingly, a reward model (RM) was trained based on the comparison dataset to predict the human-preferred output. Next, the RM was used as a reward function to fine-tune the supervised policy and maximize this reward. The test results revealed that including human feedback is a promising strategy for aligning language models with human intents. Although ChatGPT has attracted immense attention from the public, few research efforts have been made to leverage the ability of ChatGPT in construction. Moreover, how to integrate ChatGPT and CV for generating daily construction reports is not clear.

2.4. Research gaps, objectives, and scope

As reviewed in the above subsections, limited efforts have been put into construction management for automatically generating daily reports from videos. The research gaps, objectives, and scope are presented as follows.

2.4.1. Research gaps

Two significant research gaps have been identified in this research:

- An integral framework is needed for automatically generating daily construction reports from on-site videos. Existing research can extract project-related information (e.g., productivity and progress) from videos in an automated manner, whereas state-of-the-art research is still in the process of converting video footage to information. However, how to organize this fragmented project information with proper text descriptions, such as reports written by engineers, remains to be clarified. To the authors' best knowledge, no such framework has been proposed in the field of automation in construction.
- Generating engineering insights from construction videos remains challenging. In daily construction reports, engineers need to analyze daily activities/progress/safety issues and provide recommendations. This type of engineering insight requires a body of knowledge and years of professional experience in construction management.

Existing vision-based methods have insufficient reasoning ability to provide reliable insights on daily construction reports.

2.4.2. Research objectives

To fill the above research gaps, three research objectives are proposed:

- Proposing an automated framework for generating daily reports from construction videos by integrating ChatGPT and CV. The proposed framework aims to present a novel method of summarizing engineering knowledge (in the daily report format) from construction videos, which consists of two main processes: 1) converting video footage to project information via vision methods and 2) transferring information to construction engineering knowledge using ChatGPT. Because ChatGPT has been trained with extremely large amounts of text data from the Internet, it possesses both general knowledge from human beings and domain knowledge in construction management and can generate daily reports that resemble those written by professional engineers.
- Evaluating the feasibility of ChatGPT in construction management. As one of the most representative LLMs, ChatGPT has attracted keen interest from the public due to its vast domain knowledge, highly accurate responses, and high processing speed. However, the feasibility of ChatGPT in construction management remains unclear. In this research, ChatGPT is adopted for two purposes: 1) organizing project information extracted by CV into the daily report format with proper text descriptions to explain the information and 2) analyzing project information to provide professional insights and recommendations in daily reports. The authors analyzed the experimental results to explore the feasibility of using ChatGPT in construction, which can potentially benefit other areas of research adopting generative language machine intelligence for construction management applications.
- Developing a web application to implement the proposed framework and conduct field tests in a real construction project. To evaluate the proposed framework, a web application is developed with the following functions: video management (i.e., uploading, browsing, and downloading), project information visualization (i.e., activity, productivity, and progress), and daily report management (generation, browsing, and downloading). The developed application is adapted to a real construction site located in Hong Kong for a month to perform field tests. Accordingly, the contributions and limitations of this research are further clarified with the support of practical evidence.

2.4.3. Research scope

This research focuses on analyzing construction videos of workers' activities in high-rise buildings, and the generated daily reports mainly contain the project productivity information. Notably, this research proposes a general framework that can generate daily reports for multiple purposes, such as productivity analysis and safety control. Previous research has already proved that the vision-based method can be used to conduct productivity analysis and safety monitoring from videos of workers, machines, and materials. Therefore, the construction productivity case was selected in this research to validate the feasibility of the proposed framework for automated daily documentation in construction management. By customizing the vision algorithms from this research, the proposed framework can easily be adapted to generate daily reports from construction machine and material videos for other project management purposes (e.g., safety control and progress monitoring). In real construction projects, the daily report contents may change according to the different construction stages. Therefore, proposing a general framework for daily report generation is necessary, and the specific algorithms can be adjusted to meet the needs of real projects.

3. Proposed framework

The proposed framework for automatically generating daily reports from construction videos is presented in this section. First, an overview of this framework is introduced. Three main modules are then detailed, namely vision-based project information extraction, ChatGPT for daily report generation, and web application development.

3.1. Overview

Fig. 1 shows an overview of the proposed framework. It connects on-site IP cameras and streams construction site videos to the server. A web application is developed to store the streamed videos, which allows engineers to manage and browse these videos. First, the construction videos are fed into the vision-based project information extraction module. In this module, all workers are detected and tracked by indicating the workers' pixel location and identification (ID) number. The tracking results have been inputted into the activity recognition algorithm to clarify worker activities into 10 categories, and the productivity information can be analyzed based on the activity information. Meanwhile, the extracted productivity information can be visualized using the web application. The activity, productivity, and other project-related information (e.g., project name, date, and weather) are then combined to form a prompt, which refers to a discussion topic the user provides for the ChatGPT model to respond to. The designed prompt is used to trigger ChatGPT to generate daily reports from construction videos. The generated reports are stored in the server and construction engineers can manage, view, and download them for further use from the web application.

The technical innovations of the proposed framework are two-fold. First, this framework integrated CV and LLMs for construction report generation, which has not been performed in the field before. The CV can extract project-related information from videos, and LLMs understand this fragment information and transform it into a daily report, which a human can understand. As such, the generated daily reports contain both subjective project information and engineering insights. Second, a prompt is designed for instructing ChatGPT to generate construction reports. Considering that ChatGPT is a novel technology, how it can be adopted for construction documentation purposes has not been clarified. The proposed framework indicates how to instruct ChatGPT to generate construction reports, which can help other researchers in this field to efficiently incorporate LLMs in construction management applications.

3.2. Vision-based project information extraction

The main purpose of this module is to extract project productivity information from video streams. The accuracy of the vision-based monitoring directly affects the report quality because all productivity data will appear in the report. To achieve this goal, four well-adopted techniques in vision-based construction site monitoring are introduced, namely work detection, worker tracking, activity recognition, and productivity analysis.

3.2.1. Worker detection

Worker detection involves producing precise bounding boxes of construction worker objects from individual video sequences using object detection techniques. In this research, the deep learning object detector YOLO-v5 [39] was adopted due to its high processing speed and reliable performance in the field [17,40,41]. The architecture of YOLO-v5 is shown in Fig. 2, which consists of a cross-stage partial network (CSPNet), path aggregation network (PANet), and YOLO layer. CSPNet serves as the backbone of YOLO-v5, extracting features from images, whereas PANet is a feature pyramid network that helps enhance the detection performance, especially for small objects. Moreover, the YOLO layers produce three types of feature maps (18×18 , 36×36 , and $72 \times$

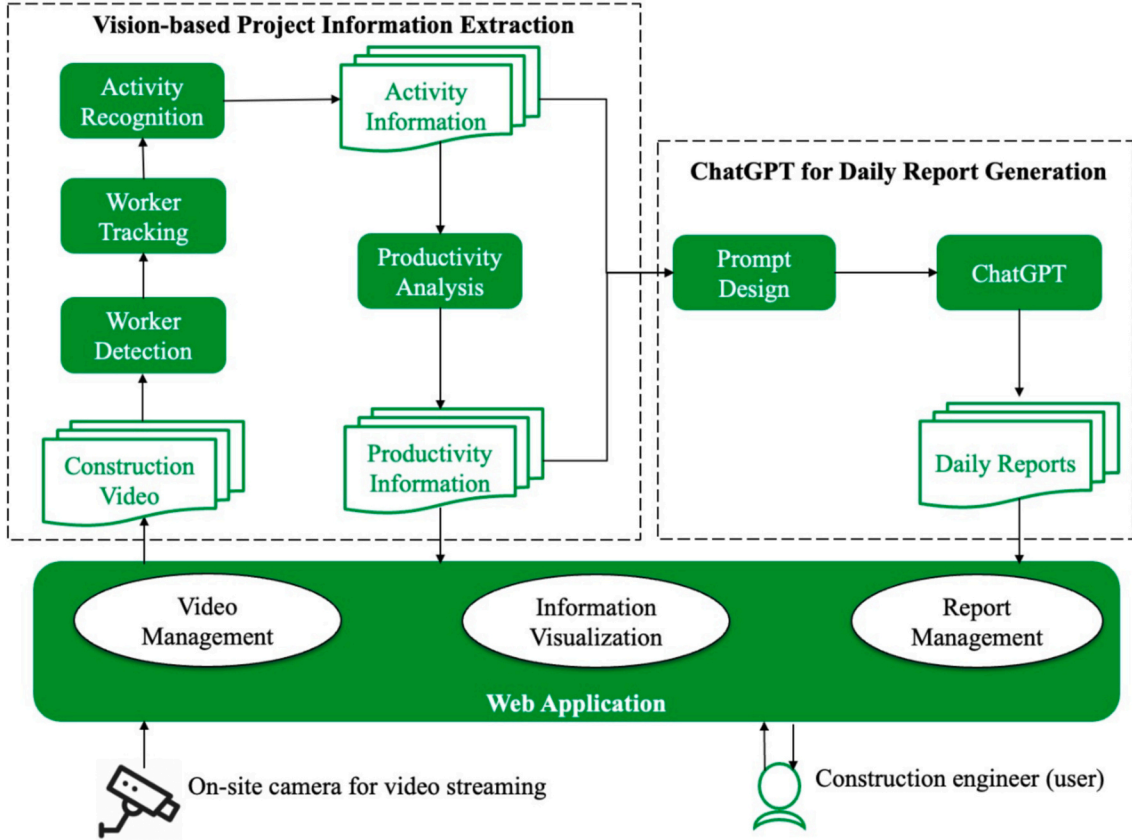


Fig. 1. Overview of the proposed framework for constructing daily report generation.

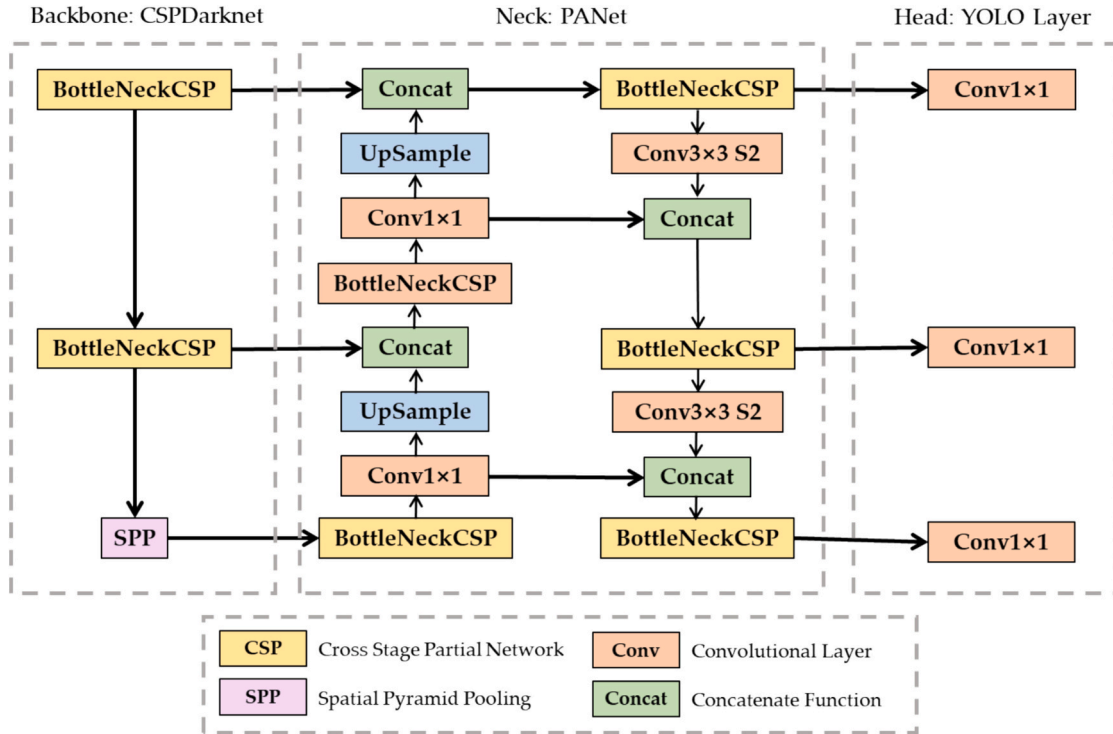


Fig. 2. Neural network architecture of YOLO-v5.

72) for multiple-scale detection to improve the detection performance of small objects. In this study, videos were captured from a camera mounted on the tower crane, and workers appeared small from a bird's eye view. The special design of YOLO-v5 helps to precisely detect small worker objects from images.

To train YOLO-v5, the authors developed an image dataset of construction workers. The developed dataset contains 1340 images, and all worker objects in each image are annotated in terms of the object category and pixel location. These images were collected from both online (e.g., YouTube and Google Image) and on-site (e.g., site visit and previous project recordings) resources. The detection results in each frame were inputted into the worker tracking algorithm.

3.2.2. Worker tracking

Worker tracking aims to retrieve the worker trajectories by associating the detection results across frames. Once all detection results are inputted into the worker tracking, the same worker object in continuous frames is assigned to the same ID number as the tracking results. The tracking results are used to identify worker activities along timelines by analyzing the worker object areas using deep neural networks. In this study, the simple online and real-time (SORT) method [42] was used for tracking multiple workers from construction videos. The performance of SORT has been validated on the multiple objects tracking benchmark (MOT16) [43] and in previous studies in the construction field [12,16,44]. In addition, SORT can achieve a tracking speed of over 200 frames per second (fps), which is important for multiple construction worker tracking.

3.2.3. Activity recognition

The activity recognition classifies the activities of each individual worker at every frame into one of the following activities: rebar, formwork, scaffold, concrete, prepare, transport, rest/talk, walk, and clean. In this study, the 3D Resnet [45] method was adopted for worker activity recognition, which is a standard CV method. 3D Resnet is a type of 3D deep neural network with a similar architecture as Resnet; where 2D neural networks are changed into 3D neural networks. Specifically, the 3D Resnet adopted the mixed convolution technique to stack the 2D and 3D convolutions in the early stage of activity recognition to improve the model performance and processing speed. For a specific frame in a construction video, all worker objects in the current frame and the same object in the previous 10 frames were stacked and inputted in the 3D Resnet model. The activity category, one of the nine activity types, was output from 3D Resnet for the worker object.

This research adopted the 3D Resnet-18 model pretrained on the Kinetics-400 dataset [46] due to its high processing speed. To fine-tune this model, 1143 video clips (3 s for each) of construction workers engaging in different activities were developed. The activity information is used for 1) further productivity analysis and 2) activity information visualization in the developed web application.

3.2.4. Productivity analysis

The productivity analysis calculates workers' productivity from the activity analysis results. Notably, this research focuses on nine activities, and they are further divided into productive activities (rebar, formwork, and concrete), semi-productive activities (scaffold, prepare, and transport), and nonproductive activities (rest/talk, walk, and clean). This study focuses on productive activities. The productivity information of interest is the accumulated working time for each activity, worker productivity, and the start and end time stamp for a project. The accumulated working time can be calculated from Eq. (1).

$$\text{accumulated working time} = \frac{\text{total number of activity frames}}{\text{frames per second}} \quad (1)$$

Worker productivity can be calculated as the accumulated working time for a specific activity multiplied by the production rate per hour (as shown in Eq. (2)).

$$\text{worker productivity} = \frac{\text{accumulated working time} \times \text{productivity rate}}{\text{hour}} \quad (2)$$

The start and end timestamps can be retrieved from the timestamps of the first worker clocking in and the last worker clocking out, respectively, in the daily video footage of a project. All detection, tracking, and activity analysis results are stored in a database, which means all productivity information can be automatically analyzed and flexibly visualized (e.g., hourly, daily, weekly, or even monthly). In this research, the relational database was adopted, and the database structure of the proposed framework is shown in Fig. 3.

3.3. ChatGPT for daily report generation

The main purpose of this module is to bridge the vision-based productivity analysis and construction report generation by leveraging the feasibility of ChatGPT. First, a prompt template was designed to incorporate the daily activity and productivity information in the construction project. By inserting the daily information into the template, a prompt can be used as the query question, which is inputted to ChatGPT using the OpenAI application programming interface (API). The querying results are then exported to the web application as construction daily reports for users to browse. Details on prompt design and ChatGPT are presented in this subsection.

3.3.1. Prompt design

A prompt refers to the information, sentences, or questions that users enter into the generative AI, and it has a large effect on the quality of the feedback received by users. In brief, prompting refers to how humans interact with and provide initial instruction to the LLM. Notably, a slight difference in prompting may cause a large difference in the quality of the LLM results, and designing a proper prompt to allow LLMs to understand engineering problems is crucial.

To meet the daily report expectations in construction management, the prompt design in this research considers three types of information. 1) Basic project information (e.g., date, site name, site address, and weather), which is an essential component of a daily construction report. 2) Daily productivity information (e.g., total productivity, top three activities, first worker start time, and last worker leave time). (In this research, the authors focused on productivity analysis; more safety information can be reported by customizing the vision-based analysis module for other studies in the field.) 3) Comparison of the analysis and summaries of the productivity compared to the historical project data. Importantly, the daily report should provide engineering insights by summarizing and identifying the reasons for any delay or under-productivity. LLMs have shown breakthrough potential to generate this kind of engineering summary, which traditional methods cannot do.

The design prompt used in this study is.

"Please write a markdown format daily report in a construction site with the following information: date is {%date%}; site name is {%site name%}; site address is {%site address%}; weather is {%weather%}; total productivity is {%total productivity%}; the first worker starts working at {%first worker start time%} and the last worker leaves the site at {%last worker leave time%}; the top three activities of workers today are {%top3 activity%} and their times are {%top3 activity time%}, respectively; today is a {%productivity level%} productive working day. Please refer to the project historical data and write a {%number of words%} word summary section to analyze the daily activity and productivity performance."

Notably, {%variable%} refers to the variable directly extracted from the productivity analysis results. Specifically, the site name and site address are predefined in the construction project. Date and weather information can be extracted from the Internet. The total productivity, first worker start time, last worker leave time, top three activities, and top three activity times can be extracted from the database because all productivity information has been stored in a database, and this information changes

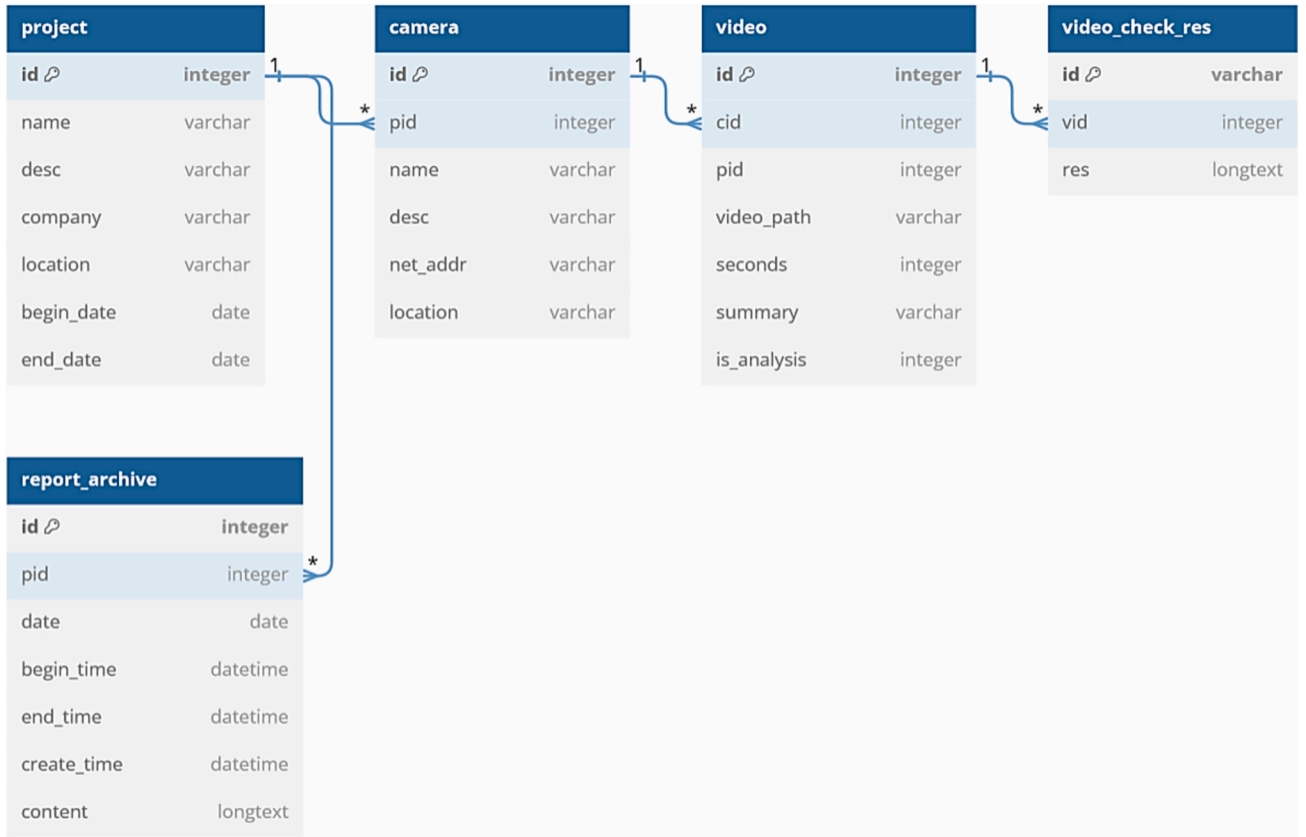


Fig. 3. Database structure of the proposed framework for productivity analysis.

every day. In this study, the daily *productivity level* was categorized as high, normal, and low. In the database, all working days were ranked in a list according to their daily productivity. If new daily data outranks the first 1/3 days in the list, the day is considered to be high-productive. Moreover, the normal and low productive levels refer to the middle 1/3 and last 1/3 sections in the list. The *number of words* is a hyper-parameter that can be determined by the users. Because LLMs can remember historical queries, they naturally know the historical productivity information of this project and can analyze the productivity performance and generate summaries without requiring additional information. In this study, the prompt will be automatically integrated to ChatGPT by: 1) extracting variable information from the database and Internet; 2) filling the variables into the aforementioned prompting template; 3) query ChatGPT with the prompt; and 4) parse the ChatGPT outputs and generate the report.

3.3.2. ChatGPT

The proposed design was directly inputted into ChatGPT through the API provided by OpenAI. The ChatGPT API provides a way to interact programmatically with the ChatGPT model hosted on OpenAI's servers. Developers can send text prompts to the API and receive responses generated by ChatGPT in real-time. This enables integration of natural language understanding and generation capabilities into applications, websites, chatbots, and other software solutions. In this study, the GPT-3.5 model was adopted by querying the API, which was trained on an immense dataset of internet-source data (570 GB of text and 176 billion parameters) to grant its great generalizability and human-like responses.

ChatGPT was adopted for generating construction reports for the following reasons: 1) It can take fragmented project information and present it in a more human-understandable way for reporting. 2) ChatGPT has been trained on an extensive amount of data, enabling it to summarize information and provide insights similar to those of construction engineers. 3) ChatGPT can retain historical input data, such as

historical productivity information, and generate summaries based on it, making it suitable for long-term project reporting spanning several months or years. Notably, ChatGPT can also customize the output format, which is beneficial for practical applications in the construction field. For example, this study queries ChatGPT as “*please write a markdown format ...*,” and ChatGPT can provide a markdown format output for easy integration with the web application.

3.4. Web application development

A web application to apply the proposed framework to real construction scenarios for efficient project management was developed in this study. The web application has three functions, and details are given as follows.

3.4.1. Video management

The video management function offers basic platforms for managing videos over the entire project lifespan. This function includes connecting IP cameras, streaming videos from IP cameras, uploading video clips from computers, organizing videos by projects/timelines, video searching by dates, video previewing, video downloading, and video deletion.

3.4.2. Project information visualization

The project information visualization function allows project managers/engineers to interact with productivity information. This function includes worker monitoring visualization, worker location visualization, activity information visualization, top three activities visualization, daily productivity visualization, and project-wise productivity visualization. All the categories of project information mentioned above can be customized for different periods to meet the demands of project managers.

3.4.3. Daily reports management

The daily reports management function allows for the management of daily reports generated by ChatGPT. This function includes daily report generation, report previewing, report storage, report download, and report deletion. By default, the web application automatically generates a report daily.

Additional information regarding the implementation of the proposed framework and web application can be found in [Section 4.1](#).

4. Implementation, experiment, and results

In this section, the implementation of the proposed system is introduced. Moreover, two experiments are designed to evaluate the CV-based productivity analysis module and the functionalities of the

proposed system and the report, respectively.

4.1. Implementation

The implementation in this study includes both server- and client-sides. Server-side means that the action takes place on a web server, which was implemented on Python 3.6. and the Flask 2.2.2. Other major libraries were also adopted for data processing, database querying, and file management, including ffmpeg-py 0.2.0, httpx 0.23.3, numpy 1.23.5, opencv-python 4.6.0.66, pdfkit 1.0.0, Pillow 9.3.0, PyJWT 2.6.0, PyMySQL 1.0.2, PyYAML 6.0, torch 1.12.1, and torchvision 0.13.1. The client-side refers to the action from the user's (the client's) computer. In this study, Javascript was used for the client-side with using following main libraries, @ant-design/icons 5.0.1, @emotion/reac 11.10.6,

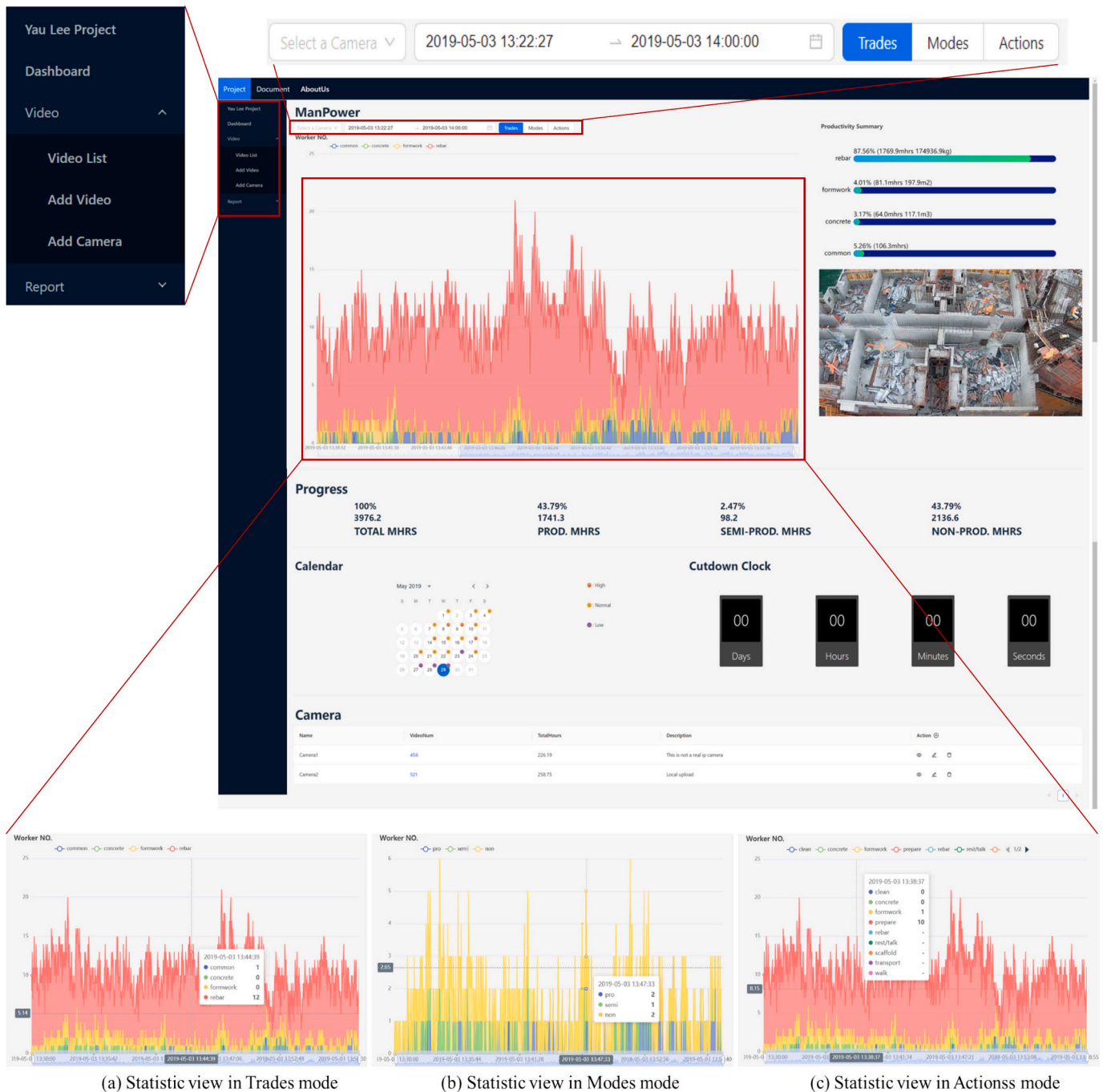


Fig. 4. UI of the main page and project information visualization function.

@emotion/styled 11.10.6, @mui/material 5.11.13, @mui/x-date-pickers 6.0.2, ahooks 3.7.5, antd 5.3.1, axios 1.3.4, dayjs 1.11.7, downloadjs 1.4.7, echarts 5.4.1, and react 18.2.0.

The user interface (UI) of the main page of the developed web application is shown in Fig. 4, which is also the project information visualization function. In the left, the project name, main dashboard, video functions, report functions are illustrated. In the Manpower section, the productivity information is visualized and can be customized by selecting a specific camera or period within the project. In our system, the statistics analysis results are presented in three types of views, namely Trades, Modes, and Actions. The Trades mode classifies worker activities based on the type of trade. In Modes, worker activities are classified as productive activities, semi-productive activities, and non-productive activities, as previously introduced. The Actions mode shows human work hours based on worker's activities. In each mode, the number of workers for each type of activity at any time point can be viewed. In addition, Fig. 5 shows the UI for the video management and daily reports management functions, which can be used to add, search, download, and delete videos/reports in a project.

4.2. Experimental design

Two experiments were conducted to evaluate the 1) vision-based productivity analysis module and 2) user experience of daily reports and web applications. The productivity analysis experiment aims to evaluate the performance of the proposed vision-based productivity analysis, whereas the user experience experiment aims to evaluate the quality of generated reports and functionalities of the developed web application. The experiments and results are presented in the following subsections.

A real construction job site was employed as the test bed, namely the Ming Wah Dai redevelopment project located in Hong Kong. Two cameras were installed at this site (on the crane to capture a bird's-eye view of the whole site) and the videos were captured from April 23 to

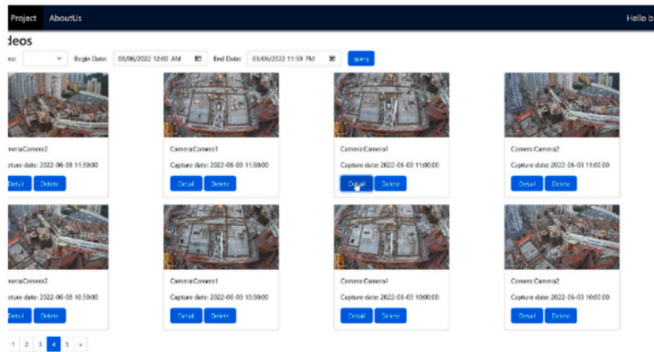
May 29, 2019 for a total of 488.5 h. To be noted, these two cameras were installed on two separated working zones and did not have overlapping views. The project mainly involved construction workers renovating a high-rise building, including rebar, formwork, and concrete. Fig. 6 shows a screenshot of daily report on 10 May 2019, which is automatically generated by the proposed framework. Specifically, the report is generated as a pdf file, which can be download from the web application.

4.3. Productivity analysis experiment and results

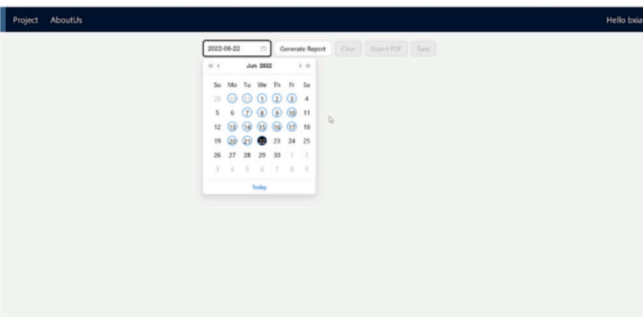
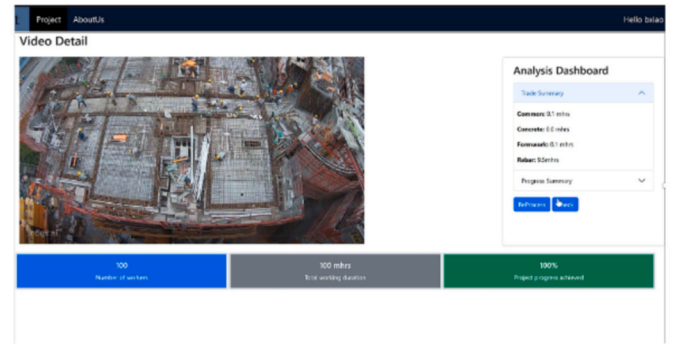
A quantitative evaluation was conducted to assess the performance of the productivity analysis. The productivity of a certain job is calculated with the duration of each activity multiplied by the production rate. The ground truth is calculated manually by observing the videos. The accuracy metric was employed to evaluate the productivity analysis performance, which is calculated from Eq. (3). Specifically, the authors selected five working days from the project videos and manually annotated the duration of three productive activities, namely rebar, formwork, and concrete. The duration of these activities in the five working days was automatically calculated using the developed vision module. Particularly, the activity duration and productivity were calculated from Eqs. (1) and (2), respectively. According to the data from Methvin, the production rates for rebar, formwork, and concrete were determined as 18.9 kg/man-hour, 1.8 m², and 1.5 m³, respectively, in this study. Notably, the production rate can be adjusted or calculated for different projects according to historical recordings, observation, or prediction.

$$Accuracy = \frac{\text{Min}\{\text{Groundtruth Productivity}, \text{Detected Productivity}\}}{\text{Max}\{\text{Groundtruth Productivity}, \text{Detected Productivity}\}} \quad (3)$$

Table 1 summarizes the experimental results for the vision-based productivity analysis module, which reveals that the accuracies for



(a) UI for video management



(a) UI for report management

Fig. 5. UI of the video and reports management functions.

Daily Report - Phase 1 Redevelopment of MING WAH DAI HA

Date: Friday, 10 May, 2019
Site Name: Phase 1 Redevelopment of MING WAH DAI HA
Site Address: 1-25A A Kung Ngam Road, Shau Kei Wan, Hong Kong
Total Productivity: 213.74 man-hours
Weather: Cloudy, 22-28 degrees

Worker Details

- The first worker started working at 08:00.
- The last worker left the site at 18:00.
- Average number of workers per hour: 353.10

Top 3 Activities

1. Rebar: 110.7 man-hours
2. Transport: 1.5 man-hours
3. Prepare: 0.5 man-hours

Summary

Today, on Friday, 10 May 2019, the construction site at Phase 1 Redevelopment of MING WAH DAI HA was operational. The site, located at 1-25A A Kung Ngam Road, Shau Kei Wan, Hong Kong, experienced a cloudy weather with temperatures ranging from 22 to 28 degrees.

A total of 213.74 man-hours were recorded for the day, with an average of 353.10 workers per hour. The first worker commenced work at 08:00, and the last worker left the site at 18:00.

The top three activities carried out by the workers today were rebar, transportation, and preparation. Rebar work accounted for the majority of the man-hours, with 110.7 hours dedicated to this task. Transportation activities took up 1.5 man-hours, while preparation activities accounted for 0.5 man-hours.

Comparing today's productivity with the project's historical data, it can be concluded that today was a normal productive working day. The productivity levels were consistent with previous records, indicating a smooth progression in the Phase 1 Redevelopment of MING WAH DAI HA project.

In conclusion, despite the cloudy weather, the construction site witnessed a productive day with a significant focus on rebar work. The average number of workers per hour remained consistent, and the overall productivity aligned with project expectations.

Fig. 6. Example of generated daily report.

Table 1
Experimental results of productivity analysis.

Activity		Rebar		Formwork		Concrete	
		Duration (Man-hour)	Productivity (kg)	Duration (Man-hour)	Productivity (m ²)	Duration (Man-hour)	Productivity (m ³)
Day1	Detected	116.85	2208.5	11.2	20.16	1.24	1.86
	Ground Truth	102.0	1927.8	10.2	18.36	1.18	1.77
	Accuracy	87.6 %		90.1 %		95 %	
Day2	Detected	120.49	2277.3	7.01	12.62	1.31	1.97
	Ground Truth	115.0	2173.5	6.0	10.8	1.41	2.12
	Accuracy	95.1 %		84.5 %		93.1 %	
Day3	Detected	52.95	1000.8	3.12	5.62	1.3	1.95
	Ground Truth	47.7	901.5	3.03	5.45	1.2	1.8
	Accuracy	90.1 %		97.1 %		92.6 %	
Day4	Detected	94.88	1793.2	10.63	19.13	12.68	19.02
	Ground Truth	90.0	1701.0	9.1	16.38	10.27	15.41
	Accuracy	95 %		85.6 %		81 %	
Day5	Detected	25.98	491.0	2.93	5.27	0.90	1.35
	Ground Truth	23.6	446.0	2.7	4.86	0.87	1.31
	Accuracy	91 %		92.1 %		97 %	
Average Accuracy		91.76 %		89.88 %		91.74 %	

rebar, formwork, and concrete were 91.76 %, 89.88 %, and 91.74 %, respectively. The average productivity accuracy in the experiment was 91.13 %, which is comparable to the current practice of construction productivity management. Notably, this study adopted the standard algorithms and pipeline used in the vision-based monitoring of construction sites and achieved similar results as previous studies, confirming that state-of-the-art vision methods can extract reliable information from videos to report generation. The experimental results also reveal the feasibility of the proposed framework by bridging CV and

ChatGPT.

4.4. User experience experiment and results

To evaluate the quality of the report and web application, 10 graduate students from the Hong Kong Polytechnic University with a background in construction management were selected to participate in the experiment. Fig. 7 shows the experiment design procedure. First, the participants were briefly trained by (1) explaining the purpose of the

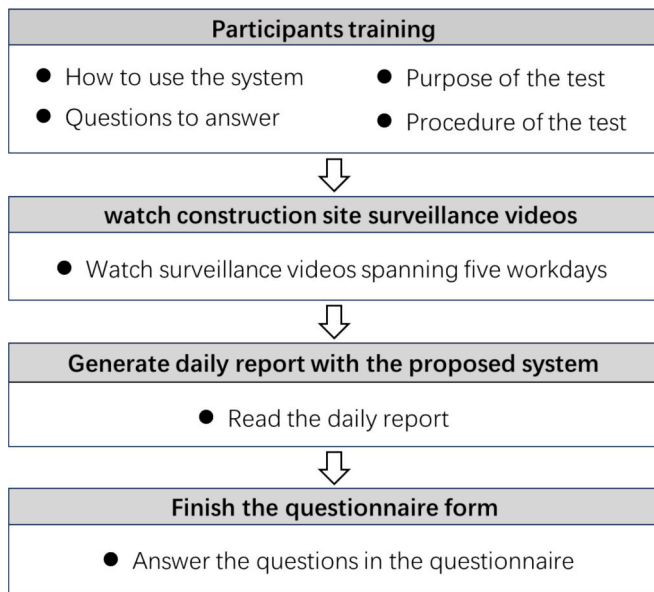


Fig. 7. Experimental procedure.

experiment, (2) indicating how to use the proposed system to generate the daily report, (3) explaining the questions on the questionnaire form, and (4) introducing the test procedure.

Moreover, based on the procedure from the training stage, each participant was asked to view the construction site surveillance videos collected from the construction site. Each participant viewed preselected videos for five random workdays and focused on the production information of the project. The videos were watched without time requirements to grant the participants the opportunity to fully observe the content. After watching each video, the participants were required to operate the developed web application and generate a report for that video. Specifically, first, they should open the web application and select the video they watched from the calendar, as shown in Fig. 4. Then, they click the button “Generate Report” (as shown in Fig. 5), and the report will be generated automatically accordingly.

Operate the developed web application and generate a report for that video. Finally, the participants were required to read generated reports and fill the questionnaire form to the best of their knowledge.

Each participant was required to express their opinion by ranking their level of agreement in the questionnaire. The questionnaire utilizes a five-level Likert scale [47]: strongly agree = 5, agree = 4, neutral = 3, disagree = 2, and strongly disagree = 1. Participants were asked to

explain the items they disagreed and strongly disagreed with and provide feedback and suggestions on the method. A total of 10 questionnaires were released and 10 were returned and judged valid for this study. The questionnaire contents and results are listed in Table 2.

The experimental results demonstrate that the overall feedback from the participants regarding the proposed system and generated report is positive, as the total score of each performance index is higher than 3.18. Among the five indexes, satisfaction with the system and the clarity of the report achieved the highest agreement among participants. Three factors achieved a high score, which indicates that participants think that the proposed system can efficiently lower manual efforts in daily documentation work, the report content is easy to understand, and the report is organized with natural language. The participants were also satisfied with the functionality of the system. The functions of the system for site visualization, report management, and video management achieved scores of 4.24, 4.28, and 4.18, respectively.

5. Discussion

5.1. Report quality analysis

After the experiment, the participants were followed up and asked to express their evaluations and provide suggestions about the report. The participants deemed the report performance satisfactory in the following aspects. The report was clearly organized in natural language, summarizing the daily work conditions of construction workers, and it can be easily understood by the construction manager. It saves human efforts associated with daily reporting. In addition, the report contains project-related construction productivity information, which is necessary and useful for daily reporting and productivity monitoring. Moreover, the information provided by the report is based on construction domain knowledge and contains engineer insight.

The scores of reliability and profession were 3.90 and 3.88, respectively. Although the scores for reliability and professionalism are both above 3, indicating that participants agree that the report can provide project-related information that is reliable and sourced from real construction sites, the scores for these two factors are relatively low compared to the other three factors. Regarding the professionalism of the report, some participants felt that the correct construction terminology was not used, and it cannot provide suggestions to improve the management of the construction. For example, one of the participants indicated that the suggestion was not detailed enough for the construction manager to perform specific productivity optimization tasks.

The quality of the proposed report can be improved in the following aspects based on users' experiences. First, the productivity information proposed by the report was too general. For example, the report only

Table 2
Experimental results of the questionnaire.

Object	Performance index	Factors	Option (number)					Total score
			Totally agree	Agree	Normal	Disagree	Strongly disagree	
System	Satisfactory	This system can reduce manual efforts in daily documentation	30	20	0	0	0	4.60
		This system can improve productivity, progress, and safety management	10	34	6	0	0	4.08
Functionality		The system provides efficient visualization information	18	26	6	0	0	4.24
		The system provides report management functions	20	25	4	1	0	4.28
		The system provides video management functions	19	21	10	0	0	4.18
Clarity		The report content is easy for humans to understand	33	16	1	0	0	4.64
		The report is organized using natural sentence descriptions	31	18	1	0	0	4.60
Reliability		The report contains project-related information required in daily reporting	13	25	6	6	0	3.90
		The report contains real information recorded from surveillance videos	13	20	15	2	0	3.88
Profession		The report uses correct construction terminology	15	14	19	2	0	3.84
		The report provides suggestions that are useful to project managers	5	21	8	10	6	3.18

indicates that the concrete work in a specific report is 6 h, making it difficult for the construction manager to understand the progress of the project. More detailed information, such as 6 h of concrete pouring work for the top floor slab was completed, would be more valuable for productivity and progress monitoring. Second, some suggestions from the report are sensible but are not applicable for improving the construction productivity. For example, the report summarizes the influence of weather conditions on the workers' productivity. However, weather condition cannot be optimized to improve the construction productivity. To better support construction management work, useful applicable information should be provided to construction managers to help them improve the productivity. For this purpose, text information related to productivity improvement strategies (i.e. a productivity improvement strategy knowledge graph) should be added into the prompt to help the ChatGPT provide suitable suggestions based on different conditions.

5.2. Prompt engineering for LLMs in construction applications

LLMs, especially ChatGPT, have exhibited immense potential to represent fragmented information as human-understandable sentences, making them suitable for various construction applications, including accidental report analysis, daily report generation, and structure inspection reporting. However, LLMs are very sensitive to prompts and some changes to the prompts may affect the querying result quality. This study draws some conclusions regarding prompt engineering. 1) Asking LLMs clear questions. A clear question helps LLMs understand the intention and give proper feedback based on their cross-disciplinary knowledge bases. For example, this study focused on generating construction daily reports and the query was "please write a markdown format daily report in a construction site with the following information: ..." 2) Providing as much project-related details as possible and combining these details using semicolons. The authors found that more project information improves the quality of generated reports. Such details should be connected using semicolons that LLMs can easily break up to understand related project information 3) LLMs can provide various output formats to accommodate project needs. For instance, the proposed framework directly asked ChatGPT to give a markdown format report because this format can be easily integrated into our web application. The authors also tried to ask ChatGPT to generate other formats of reports, which was successful without affecting the quality of the generated reports.

5.3. Reliability of ChatGPT for report generation

This research relies on ChatGPT for report generation, and the reliability of ChatGPT is critical for the overall performance. To assess the quality of the reports generated by ChatGPT, the authors have analyzed input prompts and output reports by using three metrics, which are response relevance, factual accuracy, and response completeness. The response relevance refers to grades how relevant the response context was to the question specified. The factual accuracy refers to grades whether the response generated is factually correct and grounded by the provided context, and the response completeness refers to how complete the response is as a statement. The evaluation was implemented by the UpTrain software [48], a unified platform to evaluate generative AI applications. The authors input the prompts and generated reports of five working days to the evaluation software and the results are 1.0 for response relevance, 0.8 for factual accuracy, and 1.0 response completeness on average. The factual accuracy has not achieved 1.0 because the construction site, weather, and location cannot be verified by the evaluation software. The results show the ChatGPT can generate high-quality reports to response the prompts.

5.4. Contributions and outlooks of ChatGPT in construction

The contributions of this study are three-fold. First, an automated

framework for generating daily reports from construction videos has been proposed. To the best of the authors' knowledge, efforts in this direction in the construction management field have been limited. The developed framework integrated vision-based site monitoring for extracting project information from videos and ChatGPT for generating reports (engineering knowledge) based on project information. Second, the feasibility of LLMs was validated in this study. In this research, a representative LLM, ChatGPT, was adopted for generating construction reports, and it exhibited excellent performance in organizing fragmented project information to produce documentation and provide professional insights on construction activities. Third, a web application for implementing the proposed framework was introduced and tested in a real construction site for one month. The web application successfully achieved various functions including video management, report management, and productivity analysis.

The authors believe, based on this study, that ChatGPT has great potential in construction management for documentation and communication, safety management, and the training of construction engineers. First, this study illustrates how ChatGPT can be used to facilitate documentation in a project. By leveraging the feasibility of CV and LLMs, engineers' manual documentation efforts can be lowered significantly. Moreover, ChatGPT can facilitate communication between team members and stakeholders. It can answer questions regarding project status, provide updates, and project compliance. This can help keep everyone informed and on the same page. For safety management, ChatGPT can provide immediate safety instructions to workers during their work. By integrating ChatGPT with CV, workers can understand construction hazards and know how to react. Regarding training, ChatGPT can help train new team members or provide refresher training to existing team members. It can guide best practices, safety protocols, and other important aspects of construction management.

5.5. Limitations

The proposed framework still has some limitations that should be improved in future work. First, some information provided by the report is not reliable. Initially, during the project, the historical information that can be used by ChatGPT for data analysis is relatively limited. In this condition, it may provide fake information. For example, if the daily productivity is below average or the progress is behind schedule at the beginning of the project, the report still indicates that the overall productivity aligns with expectations. Our experiments revealed that daily reporting becomes increasingly more precise as the project progresses because more historical data are available to ChatGPT. In addition, some prompting techniques can be used to mitigate the fake information from ChatGPT such as Chain of thought (CoT) prompting.

Second, the report generated by the proposed method may exist inconsistent formatting issues, while means the section titles have a certain level of randomness. This is an internal limitation of ChatGPT and maybe addressed by using a stronger or more restricted prompt for querying ChatGPT. In addition, the ChatGPT does not sensitive to the expression of word counts in some cases, which may result to misfunctionality of the "{%number of words%}" requirement in the prompt. In the future, the authors will investigate other new LLM models. To be noticed, due to the token limitation of ChatGPT, the proposed platform can only analyze historical data of 10 days. This limitation can be solved by using newer version of ChatGPT (i.e. ChatGPT 4).

Thirdly, the proposed system should also be improved to lower the learning cost. Specifically, in the experiment, all the participants were trained to use the system, which requires extra man-hours and costs. Generally, it took about half an hour to train 10 participants to use the proposed system, corresponding to 3 min per person. To help users better understand how to use the proposed system while increasing its efficiency, the system should be improved based on the users' preferences and habits. Third, the proposed system only focused on generating

a daily report of construction productivity. However, daily construction reports cover more than productivity information. In future work, daily reports on construction safety and quality control should also be incorporated into the proposed system.

6. Conclusions

This paper presented a system that combined vision and ChatGPT to automatically generate daily construction reports. The proposed system contained three major components: vision-based construction activity recognition and productivity analysis, report generation with ChatGPT, and a web application. First, Yolo-V5 worker detector, SORT tracker, and 3D Resnet activity recognition modes were used to recognize nine types of worker activities. Accordingly, the worker productivity was analyzed based on the activity information. Prompts were then designed based on productivity information and project-related information and fed into ChatGPT to generate daily reports. Moreover, a web application was developed to store the streamed site surveillance videos by integrating CV-based methods, report generation, and UI for automatic productivity report generation. Finally, a questionnaire-based experiment was conducted to validate the feasibility of the proposed system. The experimental results revealed that the functions of the proposed system scored 4.24 out of 5.

The contributions of this study are three-fold: First, a novel method that integrated vision and language methods was developed to extract engineering knowledge and automatically generate daily construction reports. The proposed method decreases construction managers' efforts of project documentation by utilizing generative AI. Second, the feasibility of utilizing LLMs, such as ChatGPT, for construction knowledge management was evaluated, which can be beneficial for adopting generative language machine intelligence for construction management applications. Third, a web application is developed to implement the proposed framework and conduct field tests in a real construction project.

Future work will focus on addressing the following issues. First, the proposed system can be improved to generate daily reports for not only productivity monitoring but also safety and quality control. For example, more visual and textual information should be added to the prompts to provide a more comprehensive and in-depth analysis of the construction process. Second, the proposed web application and UI can be optimized and developed based on users' preference and habits to improve their usability. Third, considering each construction site is unique, more real construction surveillance videos collected from different construction sites should be used for testing, thus, to evaluate the performance of the proposed method. Currently, the developed platform mainly focuses on generating report, while future works will investigate more functions, such as conversation between users/project managers with the platform once any issue is identified.

CRedit authorship contribution statement

Bo Xiao: Writing – original draft, Resources, Project administration, Methodology, Investigation, Conceptualization. **Yifan Wang:** Validation, Formal analysis. **Yongpan Zhang:** Software, Methodology. **Chen Chen:** Writing – original draft, Validation, Software, Investigation, Conceptualization. **Amos Darko:** Writing – review & editing, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- [1] B. Xiao, X. Yin, S.C. Kang, Vision-based method of automatically detecting construction video highlights by integrating machine tracking and CNN feature extraction, *Autom. Constr.* 129 (2021) 103817, <https://doi.org/10.1016/j.autcon.2021.103817>.
- [2] Y. Wang, B. Xiao, A. Bouferguene, M. Al-Hussein, H. Li, Vision-based method for semantic information extraction in construction by integrating deep learning object detection and image captioning, *Adv. Eng. Inform.* 53 (2022) 101699, <https://doi.org/10.1016/j.aei.2022.101699>.
- [3] W. Lamprey, A.R. Fayek, Developing a project status dashboard for construction project progress reporting, *Int. J. Architect. Eng. Constr.* 1 (2) (2012) 112–120, <https://doi.org/10.7492/IJAEC.2012.013>.
- [4] H. Xie, S. Fernando, S. AbouRizk, Integrating realtime project progress input into a construction simulation model, in: *Proceedings of the 2011 Winter Simulation Conference (WSC 2011)*, 2011, pp. 3443–3454, <https://doi.org/10.1109/WSC.2011.6148040>.
- [5] B. Xiao, Y. Wang, S.C. Kang, Deep learning image captioning in construction management: a feasibility study, *J. Constr. Eng. Manag.* 148 (7) (2022) 04022049, [https://doi.org/10.1061/\(ASCE\)CO.1943-7862.0002229](https://doi.org/10.1061/(ASCE)CO.1943-7862.0002229).
- [6] J.S. Bohn, J. Teizer, Benefits and barriers of construction project monitoring using high-resolution automated cameras, *J. Constr. Eng. Manag.* 136 (2010) 632–640, [https://doi.org/10.1061/\(ASCE\)CO.1943-7862.0000164](https://doi.org/10.1061/(ASCE)CO.1943-7862.0000164).
- [7] J. Yang, M.-W. Park, P.A. Vela, M. Golparvar-Fard, Construction performance monitoring via still images, time-lapse photos, and video streams: now, tomorrow, and the future, *Adv. Eng. Inform.* 29 (2015) 211–224, <https://doi.org/10.1016/j.aei.2015.01.011>.
- [8] C. Chen, B. Xiao, Y. Zhang, Z. Zhu, Automatic vision-based calculation of excavator earthmoving productivity using zero-shot learning activity recognition, *Autom. Constr.* 146 (2023) 104702, <https://doi.org/10.1016/j.autcon.2022.104702>.
- [9] H. Kim, K. Kim, H. Kim, Vision-based object-centric safety assessment using fuzzy inference: monitoring struck-by accidents with moving objects, *J. Comput. Civ. Eng.* 30 (2016) 1–13, [https://doi.org/10.1061/\(ASCE\)CP.1943-5487.0000562](https://doi.org/10.1061/(ASCE)CP.1943-5487.0000562).
- [10] Open AI ChatGPT. <https://openai.com/chatgpt/>, 2024 (accessed September 26, 2024).
- [11] T. Wu, S. He, J. Liu, S. Sun, K. Liu, Q.-L. Han, Y. Tang, A brief overview of ChatGPT: the history, status quo and potential future development, *IEEE/CAA J. Autom. Sinica* 10 (2023) 1122–1136, <https://doi.org/10.1109/JAS.2023.123618>.
- [12] B. Xiao, Q. Lin, Y. Chen, A vision-based method for automatic tracking of construction machines at nighttime based on deep learning illumination enhancement, *Autom. Constr.* 127 (2021) 103721, <https://doi.org/10.1016/j.autcon.2021.103721>.
- [13] J. Kim, S. Chi, Multi-camera vision-based productivity monitoring of earthmoving operations, *Autom. Constr.* 112 (2020) 103121, <https://doi.org/10.1016/j.autcon.2020.103121>.
- [14] C. Li, L. Ding, Falling objects detection for near miss incidents identification on construction site, in: *ASCE International Conference on Computing in Civil Engineering 2019*, 2019, pp. 138–145, <https://doi.org/10.1061/9780784482438.018>.
- [15] M. Ferdous, S.M.M. Ahsan, PPE detector: a YOLO-based architecture to detect personal protective equipment (PPE) for construction sites, *PeerJ. Comp. Sci.* 8 (2022) e999, <https://doi.org/10.7717/peerj-cs.999>.
- [16] B. Xiao, S.C. Kang, Vision-based method integrating deep learning detection for tracking multiple construction machines, *J. Comput. Civ. Eng.* 35 (2021) 04020071, [https://doi.org/10.1061/\(ASCE\)CP.1943-5487.0000957](https://doi.org/10.1061/(ASCE)CP.1943-5487.0000957).
- [17] C. Chen, Z. Zhu, A. Hammad, Automated excavators activity recognition and productivity analysis from construction site surveillance videos, *Autom. Constr.* 110 (2020) 103045, <https://doi.org/10.1016/j.autcon.2019.103045>.
- [18] J. Kim, S. Chi, Action recognition of earthmoving excavators based on sequential pattern analysis of visual features and operation cycles, *Autom. Constr.* 104 (2019) 255–264, <https://doi.org/10.1016/j.autcon.2019.03.025>.
- [19] X. Luo, H. Li, D. Cao, Y. Yu, X. Yang, T. Huang, Towards efficient and objective work sampling: recognizing workers' activities in site surveillance videos with two-stream convolutional networks, *Autom. Constr.* 94 (2018) 360–370, <https://doi.org/10.1016/j.autcon.2018.07.011>.
- [20] X. Luo, H. Li, Y. Yu, C. Zhou, D. Cao, Combining deep features and activity context to improve recognition of activities of workers in groups, *Comput. Aided Civ. Inf. Eng.* 35 (2020) 965–978, <https://doi.org/10.1111/mice.12538>.
- [21] J. Kim, S. Chi, Graph neural network-based propagation effects modeling for detecting visual relationships among construction resources, *Autom. Constr.* 141 (2022) 104443, <https://doi.org/10.1016/j.autcon.2022.104443>.
- [22] C. Chen, Z. Zhu, A. Hammad, M. Akbarzadeh, Automatic identification of idling reasons in excavation operations based on excavator–truck relationships, *J. Comput. Civ. Eng.* 35 (2021) 04021015, [https://doi.org/10.1061/\(ASCE\)CP.1943-5487.0000981](https://doi.org/10.1061/(ASCE)CP.1943-5487.0000981).
- [23] F. Ul Hassan, T. Le, X. Lv, Addressing legal and contractual matters in construction using natural language processing: a critical review, *J. Constr. Eng. Manag.* 147 (2021) 03121004, [https://doi.org/10.1061/\(ASCE\)CO.1943-7862.0002122](https://doi.org/10.1061/(ASCE)CO.1943-7862.0002122).
- [24] A.J.-P. Tixier, M.R. Halliwell, B. Rajagopalan, D. Bowman, Automated content analysis for construction safety: a natural language processing system to extract

- precursors and outcomes from unstructured injury reports, *Autom. Constr.* 62 (2016) 45–56, <https://doi.org/10.1016/j.autcon.2015.11.001>.
- [25] A. Chokor, H. Naganathan, W.K. Chong, M.E. Asmar, Analyzing Arizona OSHA injury reports using unsupervised machine learning, *Procedia Eng.* 145 (2016) 1588–1593, <https://doi.org/10.1016/j.proeng.2016.04.200>.
- [26] F. Zhang, H. Fleyeh, X. Wang, M. Lu, Construction site accident analysis using text mining and natural language processing techniques, *Autom. Constr.* 99 (2019) 238–248, <https://doi.org/10.1016/j.autcon.2018.12.016>.
- [27] X. Luo, X. Li, X. Song, Q. Liu, Convolutional neural network algorithm-based novel automatic text classification framework for construction accident reports, *J. Constr. Eng. Manag.* 149 (2023) 04023128, <https://doi.org/10.1061/JCEMD4.COENG-13523>.
- [28] D.A. Salama, N.M. El-Gohary, Automated compliance checking of construction operation plans using a deontology for the construction domain, *J. Comput. Civ. Eng.* 27 (2013) 681–698, [https://doi.org/10.1061/\(ASCE\)CP.1943-5487.0000298](https://doi.org/10.1061/(ASCE)CP.1943-5487.0000298).
- [29] J. Zhang, N.M. El-Gohary, Automated information transformation for automated regulatory compliance checking in construction, *J. Comput. Civ. Eng.* 29 (2015) B4015001, [https://doi.org/10.1061/\(ASCE\)CP.1943-5487.0000427](https://doi.org/10.1061/(ASCE)CP.1943-5487.0000427).
- [30] J. Song, J. Kim, J.-K. Lee, NLP and deep learning-based analysis of building regulations to support automated rule checking system, in: *ISARC. Proceedings of the International Symposium on Automation and Robotics in Construction 35*, 2018, pp. 1–7, in: <https://www.proquest.com/conference-papers-proceedings/nlp-deep-learning-based-analysis-building/docview/2123611147/se-2>.
- [31] B. Zhong, X. Xing, H. Luo, Q. Zhou, H. Li, T. Rose, Deep learning-based extraction of construction procedural constraints from construction regulations, *Adv. Eng. Inform.* 43 (2020) 101003, <https://doi.org/10.1016/j.aei.2019.101003>.
- [32] S. Moon, S. Chi, S.-B. Im, Automated detection of contractual risk clauses from construction specifications using bidirectional encoder representations from transformers (BERT), *Autom. Constr.* 142 (2022) 104465, <https://doi.org/10.1016/j.autcon.2022.104465>.
- [33] J. Devlin, M.-W. Chang, K. Lee, K. Toutanova, BERT: pre-training of deep bidirectional transformers for language understanding, *arXiv* (2019), doi: 10.48550/arXiv.1810.04805.
- [34] Y. Kim, S. Bang, J. Sohn, H. Kim, Question answering method for infrastructure damage information retrieval from textual data using bidirectional encoder representations from transformers, *Autom. Constr.* 134 (2022) 104061, <https://doi.org/10.1016/j.autcon.2021.104061>.
- [35] A. Radford, K. Narasimhan, T. Salimans, I. Sutskever, Improving language understanding by generative pre-training, 2024. https://cdn.openai.com/research-covers/language-unsupervised/language_understanding_paper.pdf (accessed September 26 2024).
- [36] I. Solaiman, M. Brundage, J. Clark, A. Askell, A. Herbert-Voss, J. Wu, Release strategies and the social impacts of language models, *arXiv* (2019), doi:10.48550/arXiv.1908.09203.
- [37] L. Ouyang, J. Wu, X. Jiang, D. Almeida, C.L. Wainwright, P. Mishkin, Training language models to follow instructions with human feedback, *arXiv* (2022), <https://doi.org/10.48550/arXiv.2203.02155>.
- [38] A. Radford, J. Wu, R. Child, D. Luan, D. Amodei, I. Sutskever, Language models are unsupervised multitask learners, *OpenAI blog*. 1 (8) (2018). https://ailab-ua.github.io/courses/resources/GPT-2_Radford_2018.pdf (accessed September 26, 2024).
- [39] Y. Zhang, Z. Guo, J. Wu, Y. Tian, H. Tang, X. Guo, Real-time vehicle detection based on improved yolo v5, *Sustainability* 14 (19) (2022) 12274, <https://doi.org/10.3390/su141912274>.
- [40] C. Chen, H. Gu, S. Lian, Y. Zhao, B. Xiao, Investigation of edge computing in computer vision-based construction resource detection, *Buildings* 12 (2022) 2167, <https://doi.org/10.3390/buildings12122167>.
- [41] H. Song, X. Zhang, J. Song, J. Zhao, Detection and tracking of safety helmet based on DeepSort and YOLOv5, *Multimed. Tools Appl.* 82 (7) (2023) 10781–10794, <https://doi.org/10.1007/s11042-022-13305-0>.
- [42] A. Bewley, Z. Ge, L. Ott, F. Ramos, B. Upcroft, Simple online and realtime tracking, in: *2016 IEEE International Conference on Image Processing (ICIP)*, 2016, pp. 3464–3468, <https://doi.org/10.1109/ICIP.2016.7533003>.
- [43] A. Milan, L. Leal-Taixe, I. Reid, S. Roth, K. Schindler, MOT16: a benchmark for multi-object tracking, *arXiv* (2016), <https://doi.org/10.48550/arXiv.1603.00831>.
- [44] B. Xiao, Y. Zhang, Y. Chen, X. Yin, A semi-supervised learning detection method for vision-based monitoring of construction sites by integrating teacher-student networks and data augmentation, *Adv. Eng. Inform.* 50 (2021) 101372, <https://doi.org/10.1016/j.aei.2021.101372>.
- [45] K. Hara, H. Kataoka, Y. Satoh, Can spatiotemporal 3D CNNs retrace the history of 2D CNNs and ImageNet?, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2018, pp. 6546–6555, <https://doi.org/10.48550/arXiv.1711.09577>.
- [46] W. Kay, J. Carreira, K. Simonyan, B. Zhang, C. Hillier, S. Vijayanarasimhan, The kinetics human action video dataset, *arXiv* (2017), <https://doi.org/10.48550/arXiv.1705.06950>.
- [47] A.T. Jebb, V. Ng, L. Tay, A review of key Likert scale development advances: 1995–2019, *Front. Psychol.* 12 (2021) 637547, <https://doi.org/10.3389/fpsyg.2021.637547>.
- [48] Uptrain-ai/Uptrain. <https://github.com/uptrain-ai/uptrain>, 2024 (accessed September 26, 2024).